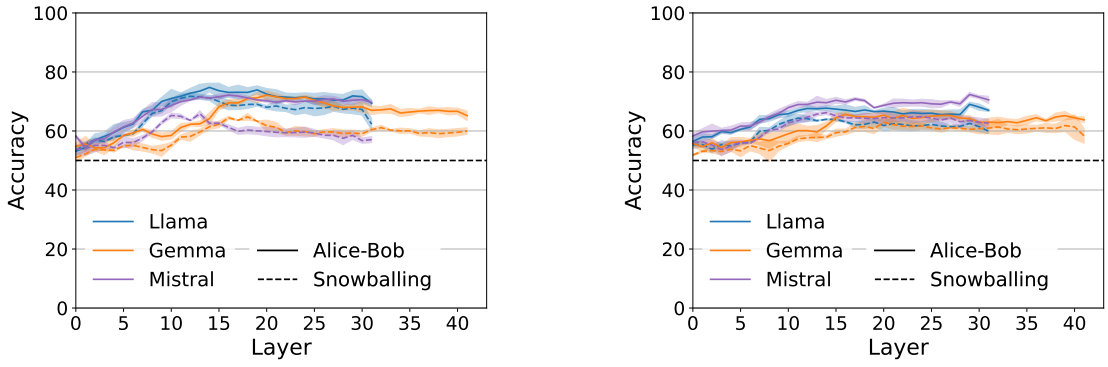


(a) TriviaQA.

(b) Natural Questions.

Figure 4: Distinguishing factually correct from HK^+, when training on examples from either a Snowballing setting or an Alice-Bob setting, and testing on the Alice-Bob setting. While the change of setting reduces accuracy, the classifier still performs substantially above a random baseline.

splitting questions the model knows how to answer into these two categories.

Figure 4 displays the results when testing on the Alice-Bob setting using classifiers trained on either examples from the same setting (solid lines) or from the Snowballing setting (dashed lines). This presents a significant challenge due to the inherent differences between the two settings. While testing on a setting different than the one used for training lowers the results, they are only lower by an average of 5% at the 15th layer and remain above the random baseline, showing some degree of generalization. (See Appendix J for similar results of generalization between other settings.) These positive results lend support to the use of such synthetic datasets for studying the phenomenon of hallucinations despite knowledge.

6 Importance of Model-specific Dataset

After demonstrating that HK^+ is a broad phenomenon and highlighting its uniqueness across different models, we now show that it can benefit from model-specific datasets. First, we compare HK^+ detection using model-specific vs. generic datasets. Next, we show the possibility of detecting HK^+ prior to generation. We argue that these results highlight the importance of model-specific datasets for HK^+.

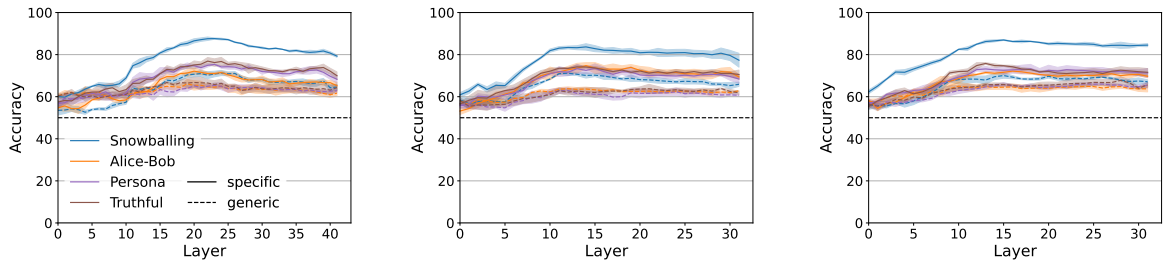
6.1 HK^+ Detections Improves with a Model-specific Dataset

In this section, we show the importance of working with a model-specific dataset instead of a generic one. We start by explaining how to construct the generic dataset and then move to the experiments.

Generic Dataset. A generic dataset is a labeled dataset that does not account for model-specific hallucinations or knowledge. Using a generic dataset is a common practice in the field of hallucination research for both detection and mitigation (e.g., Li et al. (2023); Chen et al. (2024); Zhang et al. (2024a); Marks and Tegmark (2023); Hoscilowicz et al. (2024)). The typical method for constructing a labeled CBQA dataset involves using a triplet (q, a_g, a_h), where q is a question, a_g is the gold answer, and a_h is the hallucinated answer. A hallucination example is created by concatenating a_h after q, while a factually-correct example is formed by appending a_g after q. This labeling approach is based on the correctness of the answers relative to world knowledge. However, many datasets only include the a_g answer, necessitating the creation of a_h. Following Li et al. (2023), we generate a_h by prompting an LLM (Mistral) to produce a plausible yet incorrect answer. See Appendix B for dataset creation details.

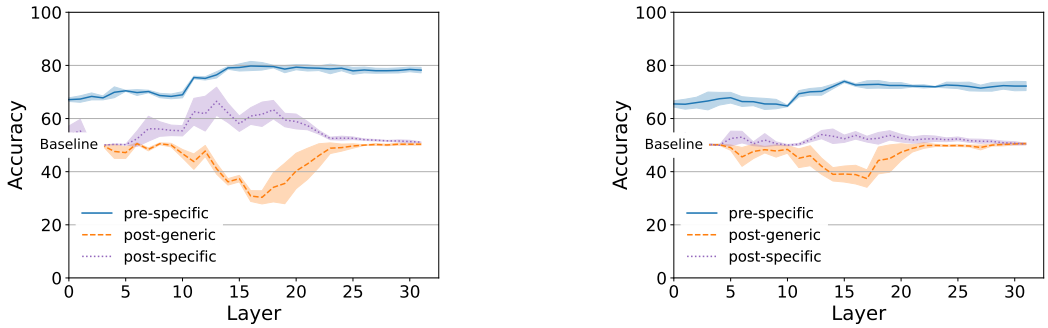
Results. As this work focuses on HK^+ cases (of hallucinations despite knowledge), we aim to show that the generic dataset is not effective in catching them. To this end, we compare classifiers trained in two binary settings: (1) model-specific setting, separating HK^+ from factually correct examples; and (2) generic setting, separating hallucinations from factually correct examples. We always test on the model-specific test set of HK^+ and factually correct examples. To make the generic and specific datasets more comparable, we add the Snowballing setting at the start of the prompt in both cases.

As Figure 5 shows, classifiers trained on hidden states obtained from generic datasets (dashed lines) demonstrate varying degrees of effectiveness and are always worse than classifiers trained on states from model-specific datasets (solid lines), which maintain relatively high accuracy. This comparison stresses the need to tailor hallucination detection to individual models, as it more effectively captures model-specific nuances and leads to more reliable identification of hallucinations per model. These results are consistent with related work (CH-Wang et al., 2023) that showed that a generic detector performs worse on specific datasets than one trained on specific datasets. However, unlike us, prior work used hallucination datasets that do not distinguish between the two hallucination types, HK^- and HK^+.



(a) Gemma. (b) Llama. (c) Mistral.

Figure 5: Distinguishing factually correct from HK^+ using classifiers trained on generic vs. model-specific datasets on TriviaQA. Accuracy with classifiers trained on specific datasets is significantly higher than generic dataset. Appendix G has similar results on Natural Questions.



(a) TriviaQA (b) Natural Questions

Figure 6: Comparing preemptive HK^+ detection using classifiers trained on model-specific pre-hallucination, generic post-hallucination, and model-specific post-hallucination examples on Mistral. Preemptive hallucination is effective using model-specific datasets, while detectors trained on generic datasets are ineffective.

6.2 Model-Specific Datasets Enable Preemptive HK^+ Detection

Our previous detection results used hidden states obtained after the model generated an answer, potentially already including a hallucination. A key advantage of model-specific datasets is their ability to detect potential hallucinations preemptively, *before they are generated*, a feature not possible with generic datasets. This section explores this capability using our WACK dataset (as before, with HK^+ and factually correct examples), where each example contains only the question q without an appended answer. As a result, the classifier is trained on the internal states of the examples at the last token of the question, rather than the last token of the answer.

Figure 6 displays preemptive hallucination detection results on the TriviaQA and Natural Questions datasets using the snowballing setting on Mistral. (See Appendix H for similar results on other models.) Model-specific preemptive detection (solid lines) demonstrates strong potential, suggesting that models can effectively anticipate hallucinations before they occur. In contrast, generic post-hallucination detection—evaluated at the last token of the answer (dashed lines)—shows random or even below random performance. This indicates that such an approach is ineffective in identifying HK^+ before their generation. Note that the generic approach can only be applied post-answer, as the hallucination signal depends on externally added annotations rather than originating from the model itself. In comparison, model-specific hallucination detection applied at the answer's last token (dotted lines), produces varied outcomes: for the TriviaQA dataset, some layers achieve detection rates of 60%–70%, while for Natural Questions, the detection rates remain low. We conclude that post-hallucination settings are not effective for preemptive hallucination detection, further highlighting the benefits of model-specific datasets.

7 Discussion and Conclusion

In this work, we demonstrated the prevalence of HK^+ across models, datasets, and settings. To induce HK^+, we developed a framework to create a labeled, model-specific dataset that categorizes outputs as HK^-, HK^+, or factually correct. We examined different configurations and observed some generalization between them, indicating that synthetic settings are effective for studying HK^+. Importantly, even mild conditions can induce HK^+, showing they manifest without flawed prompts. We highlighted the importance of distinguishing between hallucinations caused by a lack of knowledge (HK^-) and those occurring despite knowledge (HK^+) by showing that mitigation via prompting can only mit-

igate HK$^+$. We found that each model exhibits distinct knowledge and hallucination patterns, underscoring the need for model-specific datasets. Additionally, we showed that generic datasets are less effective at detecting model-specific hallucinations, and that model-specific datasets enable preemptive detection.

8 Limitations

Our work has a few limitations. While we evaluated three popular models, the patterns may differ in other ones. Additionally, we used only four settings to induce hallucinations given a model's correct knowledge; there may be many other ways to achieve similar aims. Finally, we only examined the two extremes of the knowledge spectrum and used one way of quantifying knowledge. Future work can investigate this middle region.

9 Ethics Statement

Our work provides insights into hallucinations that occur despite the model having the necessary knowledge. While this understanding could be misused to make models less reliable, our goal is to advance the study of the hallucination phenomenon to ultimately improve model reliability.

References

Amos Azaria and Tom Mitchell. 2023. The internal state of an llm knows when it's lying. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 967–976.

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, and 1 others. 2020. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901.

Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. 2022. Discovering latent knowledge in language models without supervision. In *The Eleventh International Conference on Learning Representations*.

Zouying Cao, Yifei Yang, and Hai Zhao. 2023. Autohall: Automated hallucination dataset generation for large language models. *arXiv preprint arXiv:2310.00259*.

Sky CH-Wang, Benjamin Van Durme, Jason Eisner, and Chris Kedzie. 2023. Do androids know they're only dreaming of electric sheep? *arXiv preprint arXiv:2312.17249*.

Chao Chen, Kai Liu, Ze Chen, Yi Gu, Yue Wu, Mingyuan Tao, Zhihang Fu, and Jieping Ye. 2023. Inside: Llms' internal states retain the power of hallucination detection. In *The Twelfth International Conference on Learning Representations*.

Zhongzhi Chen, Xingwu Sun, Xianfeng Jiao, Fengzong Lian, Zhanhui Kang, Di Wang, and Chengzhong Xu. 2024. Truth forest: Toward multi-scale truthfulness in large language models through intervention without tuning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 20967–20974.

Yung-Sung Chuang, Yujia Xie, Hongyin Luo, Yoon Kim, James R Glass, and Pengcheng He. 2023. Dola: Decoding by contrasting layers improves factuality in large language models. In *The Twelfth International Conference on Learning Representations*.

Shehzaad Dhuliawala, Mojtaba Komeili, Jing Xu, Roberta Raileanu, Xian Li, Asli Celikyilmaz, and Jason Weston. 2023. Chain-of-verification reduces hallucination in large language models. *ArXiv*, abs/2309.11495.

Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, and 1 others. 2024. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Yanai Elazar, Nora Kassner, Shauli Ravfogel, Abhilasha Ravichander, Ed Hovy, Hinrich Schutze, and Yoav Goldberg. 2021. Measuring and improving consistency in pretrained language models. *ArXiv*, abs/2102.01017.

Shangbin Feng, Weijia Shi, Yike Wang, Wenxuan Ding, Vidhisha Balachandran, and Yulia Tsvetkov. 2024. Don't hallucinate, abstain: Identifying llm knowledge gaps via multi-llm collaboration. *arXiv preprint arXiv:2402.00367*.

Tao Ge, Xin Chan, Xiaoyang Wang, Dian Yu, Haitao Mi, and Dong Yu. 2024. Scaling synthetic data creation with 1,000,000,000 personas. *arXiv preprint arXiv:2406.20094*.

Zorik Gekhman, Jonathan Herzig, Roee Aharoni, Chen Elkind, and Idan Szpektor. 2023. TrueTeacher: Learning factual consistency evaluation with large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 2053–2070, Singapore. Association for Computational Linguistics.

Zorik Gekhman, Gal Yona, Roee Aharoni, Matan Eyal, Amir Feder, Roi Reichart, and Jonathan Herzig. 2024. Does fine-tuning llms on new knowledge encourage hallucinations? *arXiv preprint arXiv:2405.05904*.

Jinwen He, Yujia Gong, Kai Chen, Zijin Lin, Chengan Wei, and Yue Zhao. 2023. Llm factoscope: Uncovering llms' factual discernment through intermediate data analysis. *arXiv preprint arXiv:2312.16374*.

Jakub Hoscilowicz, Adam Wiacek, Jan Chojnacki, Adam Cieslak, Leszek Michon, Vitalii Urbanevych, and Artur Janicki. 2024. Nl-iti: Optimizing probing and intervention for improvement of iti method. *arXiv preprint arXiv:2403.18680*.

Ziwei Ji, Delong Chen, Etsuko Ishii, Samuel Cahyawijaya, Yejin Bang, Bryan Wilie, and Pascale Fung. 2024. Llm internal states reveal hallucination risk faced with a query. *arXiv preprint arXiv:2407.03282*.